

KUWAIT UNIVERSITY
Department of Mathematics

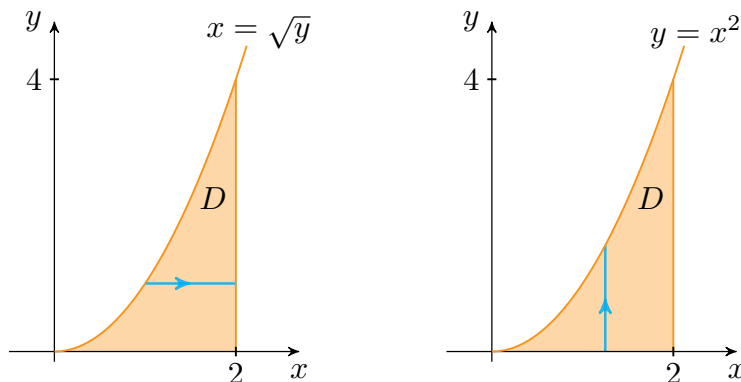
MATH 211: Calculus 3

Spring 2021–22 • Second Midterm Exam • 21 May 2022

- Q1.** [10 pts] Evaluate $\int_0^2 \int_0^\pi y^2 \sin x \, dx \, dy$.
- Q2.** [15 pts] Evaluate $\int_0^4 \int_{\sqrt{y}}^2 e^{x^3} \, dx \, dy$ by reversing the order of integration.
- Q3.** [15 pts] Use polar coordinates to find the volume of the solid that lies below the paraboloid $z = y - x^2 - y^2$ and above the xy -plane.
- Q4.** [10+5 pts] A lamina occupies the region D bounded by the parabola $y = 4 - x^2$ and the x -axis, and has density function $\rho(x, y) = y$.
- (a) Find the mass of the lamina.
 - (b) Given that $\iint_D y^2 \, dA = \frac{4096}{105}$, find the center of mass of the lamina.
- Q5.** [15 pts] Let E be the solid that is bounded by the parabolic cylinders $y = x^2$ and $x = y^2$, and the planes $z = 0$ and $z = x + y$. Suppose that f is a function that is continuous on E . Express $\iiint_E f(x, y, z) \, dV$ as an iterated integral.
- Q6.** [15 pts] Evaluate $\iiint_E z \, dV$ where E is the solid enclosed by the paraboloid $z = x^2 + y^2$ and the plane $z = 4$.
- Q7.** [15 pts] Evaluate $\iiint_E (x + z) \, dV$ where E is the solid region that lies inside the sphere $x^2 + y^2 + z^2 = 4$ and below the cone $z = \sqrt{3x^2 + 3y^2}$.

Q1. $\int_0^2 \int_0^\pi y^2 \sin x \, dx \, dy = \int_0^2 y^2 \left[-\cos x \right]_0^\pi dy = \int_0^2 2y^2 \, dy = \left[\frac{2}{3}y^3 \right]_0^2 = \frac{16}{3}.$

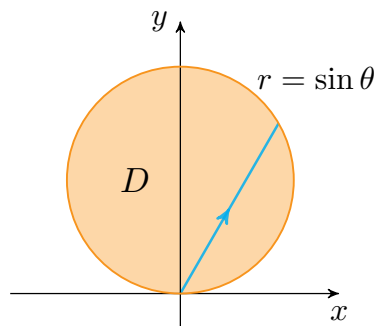
Q2. The given iterated integral $I = \int_0^4 \int_{\sqrt{y}}^2 e^{x^3} \, dx \, dy$ evaluates the double integral $\iint_D e^{x^3} \, dA$ where $D = \{(x, y) \mid 0 \leq y \leq 4, \sqrt{y} \leq x \leq 2\}$ is the Type II region sketched below.



Expressing D as a Type I region gives $D = \{(x, y) \mid 0 \leq x \leq 2, 0 \leq y \leq x^2\}$ which yields

$$I = \int_0^2 \int_0^{x^2} e^{x^3} \, dy \, dx = \int_0^2 x^2 e^{x^3} \, dx = \left[\frac{1}{3} e^{x^3} \right]_0^2 = \frac{1}{3}(e^8 - 1).$$

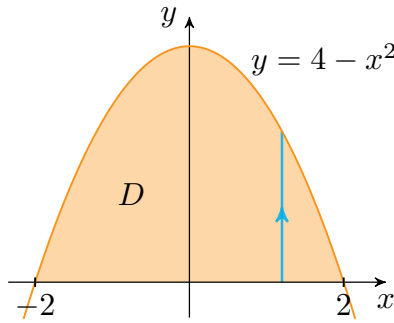
Q3. Intersecting the surfaces $z = y - x^2 - y^2$ and $z = 0$ we get $x^2 + y^2 = y$, which is the circle $r = \sin \theta$ in polar coordinates. Let D be the disk bounded by this circle.



Then the desired volume is

$$\begin{aligned} \iint_D (y - x^2 - y^2) \, dA &= \int_0^\pi \int_0^{\sin \theta} (r \sin \theta - r^2) r \, dr \, d\theta \\ &= \int_0^\pi \left[\frac{1}{3} r^3 \sin \theta - \frac{1}{4} r^4 \right]_0^{\sin \theta} d\theta = \frac{1}{12} \int_0^\pi \sin^4 \theta \, d\theta = \frac{1}{48} \int_0^\pi (1 - \cos 2\theta)^2 \, d\theta \\ &= \frac{1}{48} \int_0^\pi \left[1 - 2 \cos 2\theta + \frac{1}{2}(1 + \cos 4\theta) \right] d\theta = \frac{1}{48} \left[\frac{3}{2} \theta - \sin 2\theta + \frac{1}{8} \sin 4\theta \right]_0^\pi = \frac{\pi}{32}. \end{aligned}$$

Q4. It is evident from the illustration below that $D = \{(x, y) \mid -2 \leq x \leq 2, 0 \leq y \leq 4 - x^2\}$.



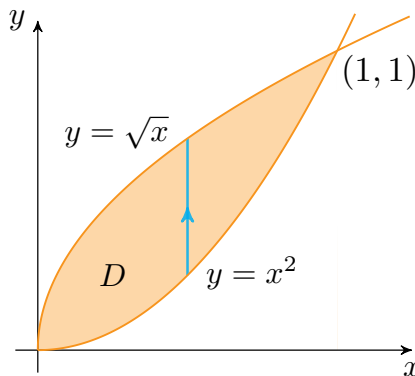
$$(a) \quad m = \iint_D \rho(x, y) dA = \int_{-2}^2 \int_0^{4-x^2} y dy dx = \int_{-2}^2 \left[\frac{1}{2} y^2 \right]_0^{4-x^2} dx = \int_{-2}^2 (16 - 8x^2 + x^4) dx = \frac{256}{15}.$$

(b) We have $\bar{x} = 0$ since the region D and the density function ρ are both symmetric with respect to the y -axis. On the other hand,

$$\bar{y} = \frac{1}{m} \iint_D y \rho(x, y) dA = \frac{15}{256} \iint_D y^2 dy dx = \frac{15}{256} \cdot \frac{4096}{105} = \frac{16}{7}.$$

We conclude that the center of mass of the given lamina is the point $\left(0, \frac{16}{7}\right)$.

Q5. The parabolic cylinders $y = x^2$ and $x = y^2$ are vertical surfaces (there is no z in their equation). Hence $E = \{(x, y, z) \mid (x, y) \in D, 0 \leq z \leq x + y\}$ where D is the region in the xy -plane that is bounded by the parabolas $y = x^2$ and $x = y^2$.



Altogether we have $E = \{(x, y, z) \mid 0 \leq x \leq 1, x^2 \leq y \leq \sqrt{x}, 0 \leq z \leq x + y\}$, which yields

$$\iiint_E f(x, y, z) dV = \int_0^1 \int_{x^2}^{\sqrt{x}} \int_0^{x+y} f(x, y, z) dz dy dx.$$

Q6. Intersecting the surfaces $z = x^2 + y^2$ and $z = 4$ we get $x^2 + y^2 = 4$. Thus

$$E = \{(x, y, z) \mid (x, y) \in D, x^2 + y^2 \leq z \leq 4\}$$

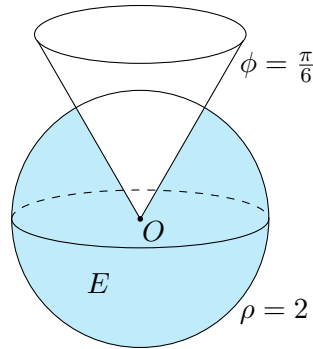
where D is the disk $x^2 + y^2 \leq 4$ in the xy -plane. In cylindrical coordinates this translates to

$$E = \{(r, \theta, z) \mid 0 \leq \theta \leq 2\pi, 0 \leq r \leq 2, r^2 \leq z \leq 4\},$$

using which we may write

$$\iiint_E z dV = \int_0^{2\pi} \int_0^2 \int_{r^2}^4 zr dz dr d\theta = (2\pi - 0) \int_0^2 r \left[\frac{1}{2} z^2 \right]_{r^2}^4 dr = \pi \int_0^2 (16r - r^5) dr = \frac{64\pi}{3}.$$

Q7. In spherical coordinates, the given sphere has equation $\rho = 2$, and the given cone has equation $\phi = \frac{\pi}{6}$. The latter is since $z = \sqrt{3}r$ gives $\tan \phi = \frac{1}{\sqrt{3}}$.



These yield

$$E = \{(\rho, \theta, \phi) \mid 0 \leq \rho \leq 2, 0 \leq \theta \leq 2\pi, \frac{\pi}{6} \leq \phi \leq \pi\},$$

and in turn,

$$\begin{aligned} \iiint_E (x + z) \, dV &= \int_{\frac{\pi}{6}}^{\pi} \int_0^{2\pi} \int_0^2 (\rho \sin \phi \cos \theta + \rho \cos \phi) \rho^2 \sin \phi \, d\rho \, d\theta \, d\phi \\ &= \int_{\frac{\pi}{6}}^{\pi} \int_0^{2\pi} (\sin^2 \phi \cos \theta + \sin \phi \cos \phi) \left[\frac{1}{4} \rho^4 \right]_0^2 \, d\theta \, d\phi \\ &= 4 \int_{\frac{\pi}{6}}^{\pi} \left[\sin^2 \phi \sin \theta + \theta \sin \phi \cos \phi \right]_0^{2\pi} \, d\phi \\ &= 8\pi \int_{\frac{\pi}{6}}^{\pi} \sin \phi \cos \phi \, d\phi = 4\pi \left[\sin^2 \phi \right]_{\frac{\pi}{6}}^{\pi} = -\pi. \end{aligned}$$